

1.1 求证: $u \times (v \times w) = (u \cdot w)v - (u \cdot v)w$

并问: $u \times (v \times w)$ 与 $(u \times v) \times w$ 是否相等? u, v, w 为矢量。

证明:

$$\begin{aligned} v \times w &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ v_x & v_y & v_z \\ w_x & w_y & w_z \end{vmatrix} \\ &= (v_y w_z - v_z w_y)\mathbf{i} + (v_z w_x - v_x w_z)\mathbf{j} + (v_x w_y - v_y w_x)\mathbf{k} \end{aligned}$$

$$\begin{aligned} u \times (v \times w) &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_x & u_y & u_z \\ v_y w_z - v_z w_y & v_z w_x - v_x w_z & v_x w_y - v_y w_x \end{vmatrix} \\ &= [u_y(v_x w_y - v_y w_x) - u_z(v_z w_x - v_x w_z)]\mathbf{i} + [u_z(v_y w_z - v_z w_y) - u_x(v_x w_y - v_y w_x)]\mathbf{j} \\ &\quad + [u_x(v_z w_x - v_x w_z) - u_y(v_y w_z - v_z w_y)]\mathbf{k} \end{aligned}$$

$$(\mathbf{u} \bullet \mathbf{w})\mathbf{v} = (u_x w_x + u_y w_y + u_z w_z)(v_x \mathbf{i} + v_y \mathbf{j} + v_z \mathbf{k})$$

$$(\mathbf{u} \bullet \mathbf{v})\mathbf{w} = (u_x v_x + u_y v_y + u_z v_z)(w_x \mathbf{i} + w_y \mathbf{j} + w_z \mathbf{k})$$

$$\begin{aligned} & (\mathbf{u} \bullet \mathbf{w})\mathbf{v} - (\mathbf{u} \bullet \mathbf{v})\mathbf{w} \\ &= (u_x w_x + u_y w_y + u_z w_z)(v_x \mathbf{i} + v_y \mathbf{j} + v_z \mathbf{k}) - \\ & \quad (u_x v_x + u_y v_y + u_z v_z)(w_x \mathbf{i} + w_y \mathbf{j} + w_z \mathbf{k}) \\ &= [u_y(v_x w_y - v_y w_x) - u_z(v_z w_x - v_x w_z)]\mathbf{i} + \\ & \quad [u_z(v_y w_z - v_z w_y) - u_x(v_x w_y - v_y w_x)]\mathbf{j} + \\ & \quad [u_x(v_z w_x - v_x w_z) - u_y(v_y w_z - v_z w_y)]\mathbf{k} \end{aligned}$$

$$\therefore \quad \mathbf{u} \times (\mathbf{v} \times \mathbf{w}) = (\mathbf{u} \bullet \mathbf{w})\mathbf{v} - (\mathbf{u} \bullet \mathbf{v})\mathbf{w}$$

$$\begin{aligned}
\mathbf{u} \times \mathbf{v} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_x & u_y & u_z \\ v_x & v_y & v_z \end{vmatrix} \\
&= (u_y v_z - u_z v_y) \mathbf{i} + (u_z v_x - u_x v_z) \mathbf{j} + (u_x v_y - u_y v_x) \mathbf{k}
\end{aligned}$$

$$\begin{aligned}
(\mathbf{u} \times \mathbf{v}) \times \mathbf{w} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_y v_z - u_z v_y & u_z v_x - u_x v_z & u_x v_y - u_y v_x \\ w_x & w_y & w_z \end{vmatrix} \\
&= [w_z (u_z v_x - u_x v_z) - w_y (u_x v_y - u_y v_x)] \mathbf{i} + [w_x (u_x v_y - u_y v_x) - w_z (u_y v_z - u_z v_y)] \mathbf{j} \\
&\quad + [w_y (u_y v_z - u_z v_y) - w_x (u_z v_x - u_x v_z)] \mathbf{k}
\end{aligned}$$

$$\therefore \quad \mathbf{u} \times (\mathbf{v} \times \mathbf{w}) \neq (\mathbf{u} \times \mathbf{v}) \times \mathbf{w}$$

1.2 求证: $(A \times B) \times (C \times D) = B(A \cdot C \times D) - A(B \cdot C \times D)$
 $= C(A \cdot B \times D) - D(A \cdot B \times C)$

证明: $A \times B = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$

$$= (A_y B_z - A_z B_y) \mathbf{i} + (A_z B_x - A_x B_z) \mathbf{j} + (A_x B_y - A_y B_x) \mathbf{k}$$

$$C \times D = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ C_x & C_y & C_z \\ D_x & D_y & D_z \end{vmatrix}$$

$$= (C_y D_z - C_z D_y) \mathbf{i} + (C_z D_x - C_x D_z) \mathbf{j} + (C_x D_y - C_y D_x) \mathbf{k}$$

$$\begin{aligned}
 \mathbf{B} \times \mathbf{C} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix} \\
 &= (B_y C_z - B_z C_y) \mathbf{i} + (B_z C_x - B_x C_z) \mathbf{j} + (B_x C_y - B_y C_x) \mathbf{k}
 \end{aligned}$$

$$\begin{aligned}
 \mathbf{B} \times \mathbf{D} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ B_x & B_y & B_z \\ D_x & D_y & D_z \end{vmatrix} \\
 &= (B_y D_z - B_z D_y) \mathbf{i} + (B_z D_x - B_x D_z) \mathbf{j} + (B_x D_y - B_y D_x) \mathbf{k}
 \end{aligned}$$

$$(\boldsymbol{A} \times \boldsymbol{B}) \times (\boldsymbol{C} \times \boldsymbol{D}) = \begin{vmatrix} \boldsymbol{i} & \boldsymbol{j} & \boldsymbol{k} \\ A_y B_z - A_z B_y & A_z B_x - A_x B_z & A_x B_y - A_y B_x \\ C_y D_z - C_z D_y & C_z D_x - C_x D_z & C_x D_y - C_y D_x \end{vmatrix}$$

$$\mathbf{A} \bullet \mathbf{C} \times \mathbf{D} = A_x(C_y D_z - C_z D_y) + A_y(C_z D_x - C_x D_z) + A_z(C_x D_y - C_y D_x)$$

$$\mathbf{B} \bullet \mathbf{C} \times \mathbf{D} = B_x(C_y D_z - C_z D_y) + B_y(C_z D_x - C_x D_z) + B_z(C_x D_y - C_y D_x)$$

$$\begin{aligned} & \mathbf{B}(\mathbf{A} \bullet \mathbf{C} \times \mathbf{D}) - \mathbf{A}(\mathbf{B} \bullet \mathbf{C} \times \mathbf{D}) \\ &= [A_x(C_y D_z - C_z D_y) + A_y(C_z D_x - C_x D_z) + A_z(C_x D_y - C_y D_x)] \cdot \\ & \quad (B_x \mathbf{i} + B_y \mathbf{j} + B_z \mathbf{k}) - \\ & \quad [B_x(C_y D_z - C_z D_y) + B_y(C_z D_x - C_x D_z) + B_z(C_x D_y - C_y D_x)] \cdot \\ & \quad (A_x \mathbf{i} + A_y \mathbf{j} + A_z \mathbf{k}) \\ &= \{B_x[A_y(C_z D_x - C_x D_z) + A_z(C_x D_y - C_y D_x)] - \\ & \quad A_x[B_y(C_z D_x - C_x D_z) + B_z(C_x D_y - C_y D_x)]\} \mathbf{i} + \\ & \quad \{B_y[A_x(C_y D_z - C_z D_y) + A_z(C_x D_y - C_y D_x)] - \\ & \quad A_y[B_x(C_y D_z - C_z D_y) + B_z(C_x D_y - C_y D_x)]\} \mathbf{j} + \\ & \quad \{B_z[A_x(C_y D_z - C_z D_y) + A_y(C_z D_x - C_x D_z)] - \\ & \quad A_z[B_x(C_y D_z - C_z D_y) + B_y(C_z D_x - C_x D_z)]\} \mathbf{k} \end{aligned}$$

$$\mathbf{A} \bullet \mathbf{B} \times \mathbf{D} = A_x(B_y D_z - B_z D_y) + A_y(B_z D_x - B_x D_z) + A_z(B_x D_y - B_y D_x)$$

$$\mathbf{A} \bullet \mathbf{B} \times \mathbf{C} = A_x(B_y C_z - B_z C_y) + A_y(B_z C_x - B_x C_z) + A_z(B_x C_y - B_y C_x)$$

$$\mathbf{C}(\mathbf{A} \bullet \mathbf{B} \times \mathbf{D}) - \mathbf{D}(\mathbf{A} \bullet \mathbf{B} \times \mathbf{C})$$

$$= [A_x(B_y D_z - B_z D_y) + A_y(B_z D_x - B_x D_z) + A_z(B_x D_y - B_y D_x)] \cdot$$

$$(C_x \mathbf{i} + C_y \mathbf{j} + C_z \mathbf{k}) -$$

$$[A_x(B_y C_z - B_z C_y) + A_y(B_z C_x - B_x C_z) + A_z(B_x C_y - B_y C_x)] \cdot$$

$$(D_x \mathbf{i} + D_y \mathbf{j} + D_z \mathbf{k})$$

$$= \{C_x [A_x(B_y D_z - B_z D_y) + B_x(A_z D_y - A_y D_z)] -$$

$$D_x [A_x(B_y C_z - B_z C_y) + B_x(A_z C_y - A_y C_z)]\} \mathbf{i} +$$

$$\{C_y [A_y(B_z D_x - B_x D_z) + B_y(A_x D_z - A_z D_x)] -$$

$$D_y [A_y(B_z C_x - B_x C_z) + B_y(A_x C_z - A_z C_x)]\} \mathbf{j} +$$

$$\{C_z [A_z(B_x D_y - B_y D_x) + B_z(A_y D_x - A_x D_y)] -$$

$$D_z [A_z(B_x C_y - B_y C_x) + B_z(A_y C_x - A_x C_y)]\} \mathbf{k}$$

1.3 求证矢量的非退化性。即：若矢量 v 与它所属的矢量空间中的任意矢量 u 都正交，即： $u \cdot v = 0$ ，则矢量 $v = 0$ 。

证明：因为 u 为任意，所以可取 u_1, u_2, u_3 ，使得

$$\det U = \begin{vmatrix} u_{1x} & u_{1y} & u_{1z} \\ u_{2x} & u_{2y} & u_{2z} \\ u_{3x} & u_{3y} & u_{3z} \end{vmatrix} \neq 0$$

由 $u \cdot v = 0$ 得

$$\begin{cases} v_x u_{1x} + v_y u_{1y} + v_z u_{1z} = 0 \\ v_x u_{2x} + v_y u_{2y} + v_z u_{2z} = 0 \\ v_x u_{3x} + v_y u_{3y} + v_z u_{3z} = 0 \end{cases}$$

因为 $\det U \neq 0$ ，所以 $v_x = v_y = v_z = 0$ 是唯一零解，即： $v = 0$ 。

1.4 已知：矢量 u, v ，求证： $|u \bullet v| \leq |u||v|$

证明： $|u \bullet v| = |u||v|\sin(u, v) \leq |u||v|$

1.5 求证： $a \times b = 0 \Leftrightarrow a, b$ 线性相关。

证明：

$$a \times b = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix}$$
$$= (a_y b_z - a_z b_y)\mathbf{i} + (a_z b_x - a_x b_z)\mathbf{j} + (a_x b_y - a_y b_x)\mathbf{k} = \mathbf{0}$$

即 $a_y b_z - a_z b_y = 0$

$$a_z b_x - a_x b_z = 0$$

$$a_x b_y - a_y b_x = 0$$

或
$$\frac{a_x}{b_x} = \frac{a_y}{b_y} = \frac{a_z}{b_z} = c$$

故
$$\mathbf{a} = c\mathbf{b}$$

即, $\mathbf{a}, \mathbf{b}$ 线性相关。

1.6 求证: $[\mathbf{a} \ \mathbf{b} \ \mathbf{c}] = 0 \Leftrightarrow \mathbf{a}, \mathbf{b}, \mathbf{c}$ 线性相关。

证明:
$$[\mathbf{a} \ \mathbf{b} \ \mathbf{c}] = \mathbf{a} \bullet (\mathbf{b} \times \mathbf{c}) = 0$$

即
$$\mathbf{a} \perp \mathbf{b} \times \mathbf{c}$$

或 $\mathbf{a}, \mathbf{b}, \mathbf{c}$ 共面。三维空间中共面的三矢量线性相关。

1.7 已知：矢量 $b=2i+j-2k$, $c=i+2j+3k$, i, j, k 为笛卡儿基；
若将 c 分解为与 b 平行的矢量及垂直于 b 的矢量 a 之和，即 $c=a+mb$ 。

求 a ; m (其中 $b \cdot a = 0$)

$$\begin{aligned}\text{解: } a &= c - mb = (i + 2j + 3k) - m(2i + j - 2k) \\ &= (1 - 2m)i + (2 - m)j + (3 + 2m)k\end{aligned}$$

$$\begin{aligned}b \cdot a &= (2i + j - 2k) \cdot [(1 - 2m)i + (2 - m)j + (3 + 2m)k] \\ &= 2(1 - 2m) + 2 - m - 2(3 + 2m) \\ &= -2 - 9m = 0\end{aligned}$$

$$m = -\frac{2}{9} \qquad a = \frac{13}{9}i + \frac{20}{9}j + \frac{23}{9}k$$

1.8 利用 $d\mathbf{r} = \mathbf{g}_i dx^i$, 证明 g_{ij} 是对称正定的。

证明: $|d\mathbf{r}|^2 = \mathbf{g}_i dx^i \bullet \mathbf{g}_j dx^j = g_{ij} dx^i dx^j = g_{ji} dx^j dx^i > 0$

即 g_{ij} 是对称正定的。

1.9 求证: 对于一组非共面的 $\mathbf{g}_i$, 存在唯一的 $\mathbf{g}^j$, $\mathbf{g}^j$ 也是非共面的。

证明: 参见: 1.2.2.4 由协变基矢量求逆变基矢量式 (1.2.17) 及式 (1.2.25) 。

1.10 已知：以 i, j, k 表示三维空间中笛卡坐标基矢量，

$$\mathbf{g}_1 = \mathbf{j} + \mathbf{k}, \quad \mathbf{g}_2 = \mathbf{i} + \mathbf{k}, \quad \mathbf{g}_3 = \mathbf{i} + \mathbf{j}$$

(1) 按公式 (1.2.17) , 求 $\mathbf{g}_1, \mathbf{g}_2, \mathbf{g}_3$ 以 i, j, k 表示的式子;

(2) 求 g_{rs} 。

解：

$$[\mathbf{g}_1 \quad \mathbf{g}_2 \quad \mathbf{g}_3] = \begin{vmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{vmatrix} = 2$$

$$\mathbf{g}_2 \times \mathbf{g}_3 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{vmatrix} = -\mathbf{i} + \mathbf{j} + \mathbf{k}$$

$$\mathbf{g}^1 = \frac{1}{2}(-\mathbf{i} + \mathbf{j} + \mathbf{k})$$

$$\mathbf{g}_3 \times \mathbf{g}_1 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix} = \mathbf{i} - \mathbf{j} + \mathbf{k}$$

$$\mathbf{g}^2 = \frac{1}{2}(\mathbf{i} - \mathbf{j} + \mathbf{k})$$

$$\mathbf{g}_1 \times \mathbf{g}_2 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{vmatrix} = \mathbf{i} + \mathbf{j} - \mathbf{k}$$

$$\mathbf{g}^3 = \frac{1}{2}(\mathbf{i} + \mathbf{j} - \mathbf{k})$$

$$g_{11} = (\mathbf{j} + \mathbf{k}) \bullet (\mathbf{j} + \mathbf{k}) = 2$$

$$g_{12} = g_{21} = (\mathbf{j} + \mathbf{k}) \bullet (\mathbf{i} + \mathbf{k}) = 1$$

$$g_{13} = g_{31} = (\mathbf{j} + \mathbf{k}) \bullet (\mathbf{i} + \mathbf{j}) = 1$$

$$g_{22} = (\mathbf{i} + \mathbf{k}) \bullet (\mathbf{i} + \mathbf{k}) = 2$$

$$g_{23} = g_{32} = (\mathbf{i} + \mathbf{k}) \bullet (\mathbf{i} + \mathbf{j}) = 1$$

$$g_{33} = (\mathbf{i} + \mathbf{j}) \bullet (\mathbf{i} + \mathbf{j}) = 2$$

1.11 根据上题结果验算公式: $\mathbf{g}_j = g_{ji} \mathbf{g}^i$

解:
$$\begin{aligned}\mathbf{g}_1 &= g_{11} \mathbf{g}^1 + g_{12} \mathbf{g}^2 + g_{13} \mathbf{g}^3 \\ &= 2 \times \frac{1}{2} (-\mathbf{i} + \mathbf{j} + \mathbf{k}) + \frac{1}{2} (\mathbf{i} - \mathbf{j} + \mathbf{k}) + \frac{1}{2} (\mathbf{i} + \mathbf{j} - \mathbf{k}) \\ &= \mathbf{j} + \mathbf{k}\end{aligned}$$

$$\begin{aligned}\mathbf{g}_2 &= g_{21} \mathbf{g}^1 + g_{22} \mathbf{g}^2 + g_{23} \mathbf{g}^3 \\ &= \frac{1}{2} (-\mathbf{i} + \mathbf{j} + \mathbf{k}) + 2 \times \frac{1}{2} (\mathbf{i} - \mathbf{j} + \mathbf{k}) + \frac{1}{2} (\mathbf{i} + \mathbf{j} - \mathbf{k}) \\ &= \mathbf{i} + \mathbf{k}\end{aligned}$$

$$\begin{aligned}\mathbf{g}_3 &= g_{31} \mathbf{g}^1 + g_{32} \mathbf{g}^2 + g_{33} \mathbf{g}^3 \\ &= \frac{1}{2} (-\mathbf{i} + \mathbf{j} + \mathbf{k}) + \frac{1}{2} (\mathbf{i} - \mathbf{j} + \mathbf{k}) + 2 \times \frac{1}{2} (\mathbf{i} + \mathbf{j} - \mathbf{k}) \\ &= \mathbf{i} + \mathbf{j}\end{aligned}$$

1.12 已知: $u=2g_1+3g_2-g_3$, $v=g_1-g_2+g_3$, 基矢量同上题。运用 1.11 题求得的 g_{rs} 计算:

(1) $u \cdot v$; (2) u, v 的协变分量。

解:
$$\begin{aligned} u \bullet v &= (2g_1 + 3g_2 - g_3) \bullet (g_1 - g_2 + g_3) \\ &= 2(g_{11} - g_{12} + g_{13}) + 3(g_{21} - g_{22} + g_{23}) - g_{31} + g_{32} - g_{33} \\ &= 2(2 - 1 + 1) + 3(1 - 2 + 1) - 1 + 1 - 2 \\ &= 2 \end{aligned}$$

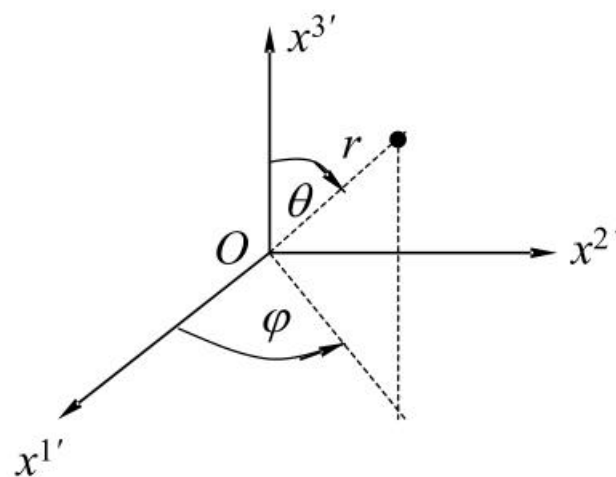
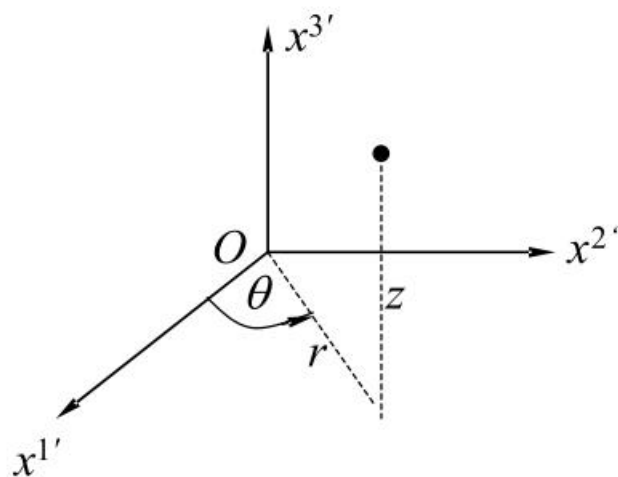
$$u_i = u^j g_{ij}, \quad v_i = v^j g_{ij}$$

$$\begin{Bmatrix} u_1 \\ u_2 \\ u_3 \end{Bmatrix} = \begin{bmatrix} g_{11} & g_{12} & g_{13} \\ g_{21} & g_{22} & g_{23} \\ g_{31} & g_{32} & g_{33} \end{bmatrix} \begin{Bmatrix} u^1 \\ u^2 \\ u^3 \end{Bmatrix} = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix} \begin{Bmatrix} 2 \\ 3 \\ -1 \end{Bmatrix} = \begin{Bmatrix} 6 \\ 7 \\ 3 \end{Bmatrix}$$

$$\begin{Bmatrix} v_1 \\ v_2 \\ v_3 \end{Bmatrix} = \begin{bmatrix} g_{11} & g_{12} & g_{13} \\ g_{21} & g_{22} & g_{23} \\ g_{31} & g_{32} & g_{33} \end{bmatrix} \begin{Bmatrix} v^1 \\ v^2 \\ v^3 \end{Bmatrix} = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix} \begin{Bmatrix} 1 \\ -1 \\ 1 \end{Bmatrix} = \begin{Bmatrix} 2 \\ 0 \\ 2 \end{Bmatrix}$$

$$\boldsymbol{u} \bullet \boldsymbol{v} = \begin{cases} u_i v^i = [6 & 7 & 3] \begin{Bmatrix} 1 \\ -1 \\ 1 \end{Bmatrix} = 2 \\ u^i v_i = [2 & 3 & -1] \begin{Bmatrix} 2 \\ 0 \\ 2 \end{Bmatrix} = 2 \end{cases}$$

- 1.13 已知：(1) 圆柱坐标系如图 (a) , $r=x^1$, $\theta=x^2$, $z=x^3$ 。
 (2) 球坐标系如图 (b) , $r=x^1$, $\theta=x^2$, $\varphi=x^3$ 。



求：两种坐标系中：

- (1) g_i 通过笛卡儿基 i, j, k 的表达式，画出简图。
- (2) 求 g^i ，说明 g^i 和 g_i 的大小与方向有何关系。
- (3) 由 g_i 求 g_{ij} , g^{ij} , $|dr|^2$ 。

解： (1)

$$\mathbf{g}_1 = \frac{\partial x^{1'}}{\partial x^1} \mathbf{i} + \frac{\partial x^{2'}}{\partial x^1} \mathbf{j} + \frac{\partial x^{3'}}{\partial x^1} \mathbf{k}$$

$$\mathbf{g}_2 = \frac{\partial x^{1'}}{\partial x^2} \mathbf{i} + \frac{\partial x^{2'}}{\partial x^2} \mathbf{j} + \frac{\partial x^{3'}}{\partial x^2} \mathbf{k}$$

$$\mathbf{g}_3 = \frac{\partial x^{1'}}{\partial x^3} \mathbf{i} + \frac{\partial x^{2'}}{\partial x^3} \mathbf{j} + \frac{\partial x^{3'}}{\partial x^3} \mathbf{k}$$

圆柱坐标系：

$$x^{1'} = r \cos \theta = x^1 \cos x^2$$

$$x^{2'} = r \sin \theta = x^1 \sin x^2$$

$$x^{3'} = z = x^3$$

$$\mathbf{g}_1 = \cos x^2 \mathbf{i} + \sin x^2 \mathbf{j}$$

$$\mathbf{g}_2 = -x^1 \sin x^2 \mathbf{i} + x^1 \cos x^2 \mathbf{j}$$

$$\mathbf{g}_3 = \mathbf{k}$$

球坐标系: $x^1 = r \sin \theta \cos \varphi = x^1 \sin x^2 \cos x^3$

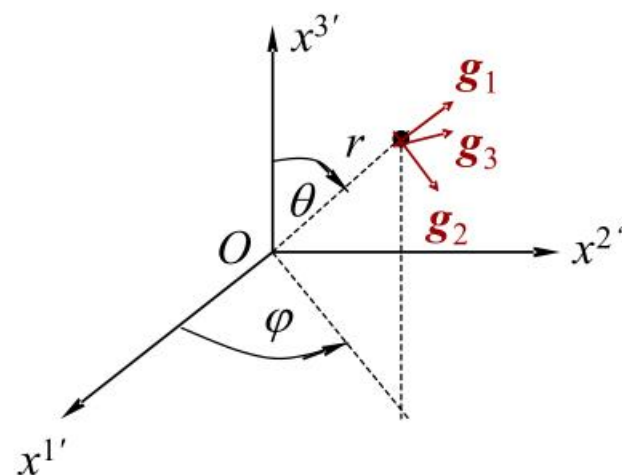
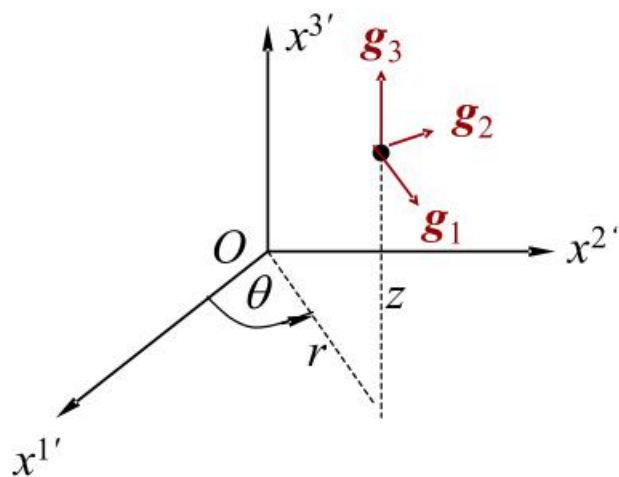
$$x^2 = r \sin \theta \sin \varphi = x^1 \sin x^2 \sin x^3$$

$$x^3 = r \cos \theta = x^1 \cos x^2$$

$$\mathbf{g}_1 = \sin x^2 \cos x^3 \mathbf{i} + \sin x^2 \sin x^3 \mathbf{j} + \cos x^2 \mathbf{k}$$

$$\mathbf{g}_2 = x^1 \cos x^2 \cos x^3 \mathbf{i} + x^1 \cos x^2 \sin x^3 \mathbf{j} - x^1 \sin x^2 \mathbf{k}$$

$$\mathbf{g}_3 = -x^1 \sin x^2 \sin x^3 \mathbf{i} + x^1 \sin x^2 \cos x^3 \mathbf{j}$$



$$(2) \quad \mathbf{g}^1 = \frac{\mathbf{g}_2 \times \mathbf{g}_3}{[\mathbf{g}_1 \quad \mathbf{g}_2 \quad \mathbf{g}_3]}, \quad \mathbf{g}^2 = \frac{\mathbf{g}_3 \times \mathbf{g}_1}{[\mathbf{g}_1 \quad \mathbf{g}_2 \quad \mathbf{g}_3]}, \quad \mathbf{g}^3 = \frac{\mathbf{g}_1 \times \mathbf{g}_2}{[\mathbf{g}_1 \quad \mathbf{g}_2 \quad \mathbf{g}_3]}$$

圆柱坐标系:
$$\begin{aligned} [\mathbf{g}_1 \quad \mathbf{g}_2 \quad \mathbf{g}_3] &= (\mathbf{g}_1 \times \mathbf{g}_2) \bullet \mathbf{g}_3 \\ &= [(\cos \theta \mathbf{i} + \sin \theta \mathbf{j}) \times (-r \sin \theta \mathbf{i} + r \cos \theta \mathbf{j})] \bullet \mathbf{k} \\ &= (r \cos^2 \theta \mathbf{k} + r \sin^2 \theta \mathbf{k}) \bullet \mathbf{k} = r \end{aligned}$$

$$\mathbf{g}^1 = \frac{1}{r} (-r \sin \theta \mathbf{i} + r \cos \theta \mathbf{j}) \times \mathbf{k} = \sin \theta \mathbf{j} + \cos \theta \mathbf{i} = \mathbf{g}_1$$

$$\mathbf{g}^2 = \frac{1}{r} \mathbf{k} \times (\cos \theta \mathbf{i} + \sin \theta \mathbf{j}) = \frac{1}{r} (\cos \theta \mathbf{j} - \sin \theta \mathbf{i}) = \frac{1}{r^2} \mathbf{g}_2$$

$$\begin{aligned} \mathbf{g}^3 &= \frac{1}{r} (\cos \theta \mathbf{i} + \sin \theta \mathbf{j}) \times (-r \sin \theta \mathbf{i} + r \cos \theta \mathbf{j}) \\ &= \cos^2 \theta \mathbf{k} + \sin^2 \theta \mathbf{k} = \mathbf{k} = \mathbf{g}_3 \end{aligned}$$

球坐标系：

$$\begin{aligned} [\mathbf{g}_1 \quad \mathbf{g}_2 \quad \mathbf{g}_3] &= (\mathbf{g}_1 \times \mathbf{g}_2) \bullet \mathbf{g}_3 \\ &= [(\sin \theta \cos \varphi \mathbf{i} + \sin \theta \sin \varphi \mathbf{j} + \cos \theta \mathbf{k}) \\ &\quad \times (r \cos \theta \cos \varphi \mathbf{i} + r \cos \theta \sin \varphi \mathbf{j} - r \sin \theta \mathbf{k})] \\ &\quad \bullet (-r \sin \theta \sin \varphi \mathbf{i} + r \sin \theta \cos \varphi \mathbf{j}) \\ &= (r \sin \theta \cos \theta \sin \varphi \cos \varphi \mathbf{k} + r \sin^2 \theta \cos \varphi \mathbf{j} \\ &\quad - r \sin \theta \cos \theta \sin \varphi \cos \varphi \mathbf{k} - r \sin^2 \theta \sin \varphi \mathbf{i} \\ &\quad + r \cos^2 \theta \cos \varphi \mathbf{j} - r \cos^2 \theta \sin \varphi \mathbf{i}) \\ &\quad \bullet (-r \sin \theta \sin \varphi \mathbf{i} + r \sin \theta \cos \varphi \mathbf{j}) \\ &= (-r \sin \varphi \mathbf{i} + r \cos \varphi \mathbf{j}) \bullet (-r \sin \theta \sin \varphi \mathbf{i} + r \sin \theta \cos \varphi \mathbf{j}) \\ &= r^2 \sin \theta \sin^2 \varphi + r^2 \sin \theta \cos^2 \varphi \\ &= r^2 \sin \theta \end{aligned}$$

$$\begin{aligned}
\mathbf{g}^1 &= \frac{1}{r^2 \sin \theta} (r \cos \theta \cos \varphi \mathbf{i} + r \cos \theta \sin \varphi \mathbf{j} - r \sin \theta \mathbf{k}) \\
&\quad \times (-r \sin \theta \sin \varphi \mathbf{i} + r \sin \theta \cos \varphi \mathbf{j}) \\
&= \frac{1}{\sin \theta} (\sin \theta \cos \theta \cos^2 \varphi \mathbf{k} + \sin \theta \cos \theta \sin^2 \varphi \mathbf{k} \\
&\quad + \sin^2 \theta \sin \varphi \mathbf{j} + \sin^2 \theta \cos \varphi \mathbf{i}) \\
&= \sin \theta \cos \varphi \mathbf{i} + \sin \theta \sin \varphi \mathbf{j} + \cos \theta \mathbf{k} = \mathbf{g}_1
\end{aligned}$$

$$\begin{aligned}
\mathbf{g}^2 &= \frac{1}{r^2 \sin \theta} (-r \sin \theta \sin \varphi \mathbf{i} + r \sin \theta \cos \varphi \mathbf{j}) \\
&\quad \times (\sin \theta \cos \varphi \mathbf{i} + \sin \theta \sin \varphi \mathbf{j} + \cos \theta \mathbf{k}) \\
&= \frac{1}{r \sin \theta} (-\sin^2 \theta \sin^2 \varphi \mathbf{k} + \sin \theta \cos \theta \sin \varphi \mathbf{j} \\
&\quad - \sin^2 \theta \cos^2 \varphi \mathbf{k} + \sin \theta \cos \theta \cos \varphi \mathbf{i}) \\
&= \frac{1}{r} (\cos \theta \cos \varphi \mathbf{i} + \cos \theta \sin \varphi \mathbf{j} - \sin \theta \mathbf{k}) = \frac{1}{r^2} \mathbf{g}_2
\end{aligned}$$

$$\begin{aligned}
\mathbf{g}^3 &= \frac{1}{r^2 \sin \theta} (-r \sin \varphi \mathbf{i} + r \cos \varphi \mathbf{j}) \\
&= \frac{1}{r^2 \sin^2 \theta} (-r \sin \theta \sin \varphi \mathbf{i} + r \sin \theta \cos \varphi \mathbf{j}) \\
&= \frac{1}{r^2 \sin^2 \theta} \mathbf{g}_3
\end{aligned}$$

$$\begin{aligned}
(3) \quad [g_{ij}] &= \begin{bmatrix} \mathbf{g}_1 \bullet \mathbf{g}_1 & \mathbf{g}_1 \bullet \mathbf{g}_2 & \mathbf{g}_1 \bullet \mathbf{g}_3 \\ \mathbf{g}_2 \bullet \mathbf{g}_1 & \mathbf{g}_2 \bullet \mathbf{g}_2 & \mathbf{g}_2 \bullet \mathbf{g}_3 \\ \mathbf{g}_3 \bullet \mathbf{g}_1 & \mathbf{g}_3 \bullet \mathbf{g}_2 & \mathbf{g}_3 \bullet \mathbf{g}_3 \end{bmatrix} \\
[g^{ij}] &= \begin{bmatrix} \mathbf{g}^1 \bullet \mathbf{g}^1 & \mathbf{g}^1 \bullet \mathbf{g}^2 & \mathbf{g}^1 \bullet \mathbf{g}^3 \\ \mathbf{g}^2 \bullet \mathbf{g}^1 & \mathbf{g}^2 \bullet \mathbf{g}^2 & \mathbf{g}^2 \bullet \mathbf{g}^3 \\ \mathbf{g}^3 \bullet \mathbf{g}^1 & \mathbf{g}^3 \bullet \mathbf{g}^2 & \mathbf{g}^3 \bullet \mathbf{g}^3 \end{bmatrix}
\end{aligned}$$

$$|\mathrm{d}\mathbf{r}|^2 = \mathrm{d}\mathbf{r} \bullet \mathrm{d}\mathbf{r} = g_i \mathrm{d}x^i \bullet g_j \mathrm{d}x^j = g_{ij} \mathrm{d}x^i \mathrm{d}x^j$$

圆柱坐标系：

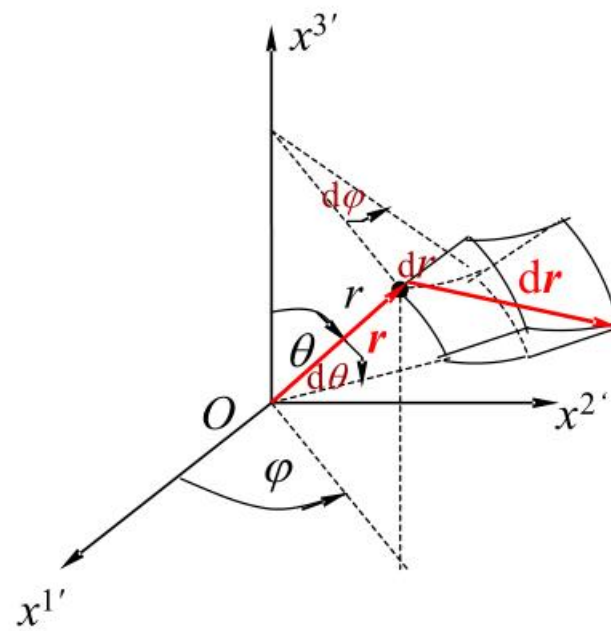
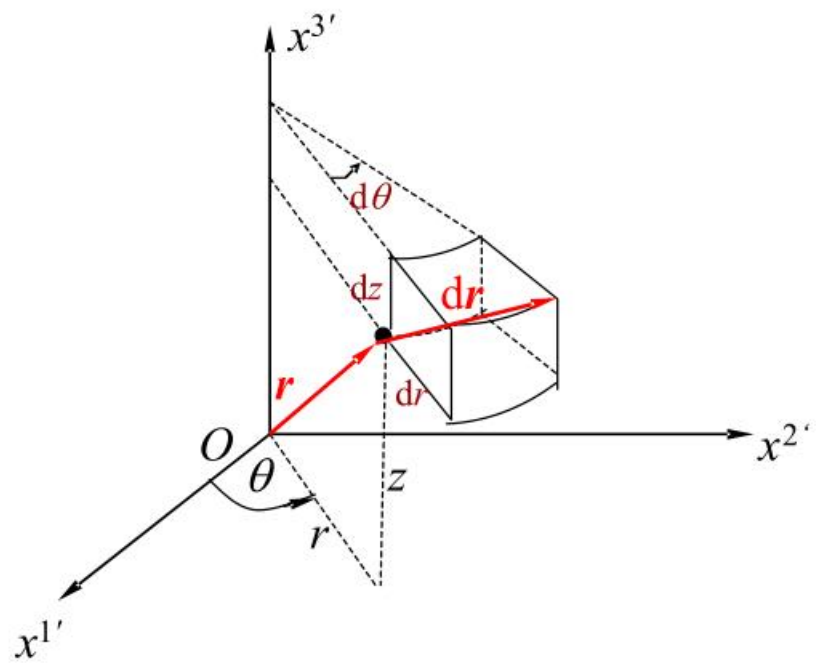
$$[g_{ij}] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad [g^{ij}] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{r^2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{aligned} |\mathbf{dr}|^2 &= [dr \quad d\theta \quad dz] \begin{bmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} dr \\ d\theta \\ dz \end{Bmatrix} \\ &= (dr)^2 + r^2 (d\theta)^2 + (dz)^2 \end{aligned}$$

球坐标系：

$$[g_{ij}] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & r^2 \sin^2 \theta \end{bmatrix}, \quad [g^{ij}] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{r^2} & 0 \\ 0 & 0 & \frac{1}{r^2 \sin^2 \theta} \end{bmatrix}$$

$$\begin{aligned} |\mathrm{d}\mathbf{r}|^2 &= [\mathrm{d}r \quad \mathrm{d}\theta \quad \mathrm{d}\varphi] \begin{bmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & r^2 \sin^2 \theta \end{bmatrix} \begin{Bmatrix} \mathrm{d}r \\ \mathrm{d}\theta \\ \mathrm{d}\varphi \end{Bmatrix} \\ &= (\mathrm{d}r)^2 + r^2 (\mathrm{d}\theta)^2 + r^2 \sin^2 \theta (\mathrm{d}\varphi)^2 \end{aligned}$$



1.14 斜圆锥面上坐标系 $x^1=\theta$, $x^2=z$,
 R, H, C 为已知 (如图)。
 求: $g_\alpha, g_{\alpha\beta}, g^\beta$ ($\alpha, \beta=1, 2$)。

解: 动点所在圆周的半径为

$$R_1 = \frac{z}{H} R$$

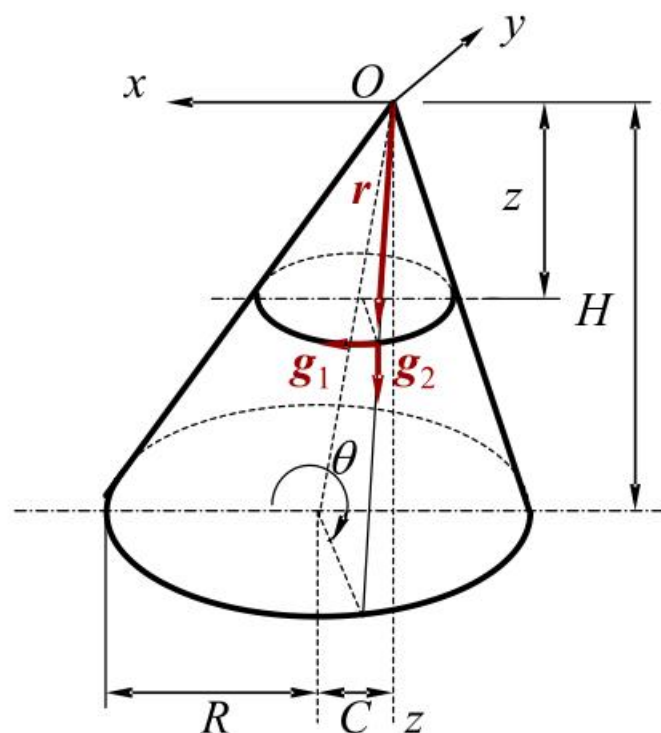
圆心至 z 轴的距离

$$C_1 = \frac{z}{H} C$$

在 xy 平面上, 该圆以 θ 为参数的方程为

$$x = C_1 + R_1 \cos \theta$$

$$y = R_1 \sin \theta$$



于是，动点的矢径为

$$\mathbf{r} = \frac{x^2}{H} (C + R \cos x^1) \mathbf{i} + \frac{x^2}{H} R \sin x^1 \mathbf{j} + x^2 \mathbf{k}$$

$$\begin{aligned} \begin{Bmatrix} \mathbf{g}_1 \\ \mathbf{g}_2 \end{Bmatrix} &= \begin{bmatrix} \frac{\partial x^{1'}}{\partial x^1} & \frac{\partial x^{2'}}{\partial x^1} & \frac{\partial x^{3'}}{\partial x^1} \\ \frac{\partial x^{1'}}{\partial x^{1'}} & \frac{\partial x^{2'}}{\partial x^{1'}} & \frac{\partial x^{3'}}{\partial x^{1'}} \\ \frac{\partial x^{1'}}{\partial x^2} & \frac{\partial x^{2'}}{\partial x^2} & \frac{\partial x^{3'}}{\partial x^2} \end{bmatrix} \begin{Bmatrix} \mathbf{i} \\ \mathbf{j} \\ \mathbf{k} \end{Bmatrix} \\ &= \frac{1}{H} \begin{bmatrix} -Rz \sin \theta & Rz \cos \theta & 0 \\ C + R \cos \theta & R \sin \theta & H \end{bmatrix} \begin{Bmatrix} \mathbf{i} \\ \mathbf{j} \\ \mathbf{k} \end{Bmatrix} \end{aligned}$$

$$[g_{\alpha\beta}] = [\mathbf{g}_\alpha \bullet \mathbf{g}_\beta]$$

$$= \frac{1}{H^2} \begin{bmatrix} -Rz \sin \theta & Rz \cos \theta & 0 \\ C + R \cos \theta & R \sin \theta & H \end{bmatrix} \begin{bmatrix} -Rz \sin \theta & C + R \cos \theta \\ Rz \cos \theta & R \sin \theta \\ 0 & H \end{bmatrix}$$

$$= \frac{1}{H^2} \begin{bmatrix} R^2 z^2 & -RCz \sin \theta \\ -RCz \sin \theta & H^2 + R^2 + C^2 + 2RC \cos \theta \end{bmatrix}$$

$$[g^{\alpha\beta}] = [g_{\alpha\beta}]^{-1}$$

$$= \frac{H^2}{R^2 [H^2 + (R + C \cos \theta)^2]} z^2 \begin{bmatrix} H^2 + R^2 + C^2 + 2RC \cos \theta & RCz \sin \theta \\ RCz \sin \theta & R^2 z^2 \end{bmatrix}$$

$$\{\mathbf{g}^\beta\} = [g^{\alpha\beta}] \{\mathbf{g}_\alpha\}$$

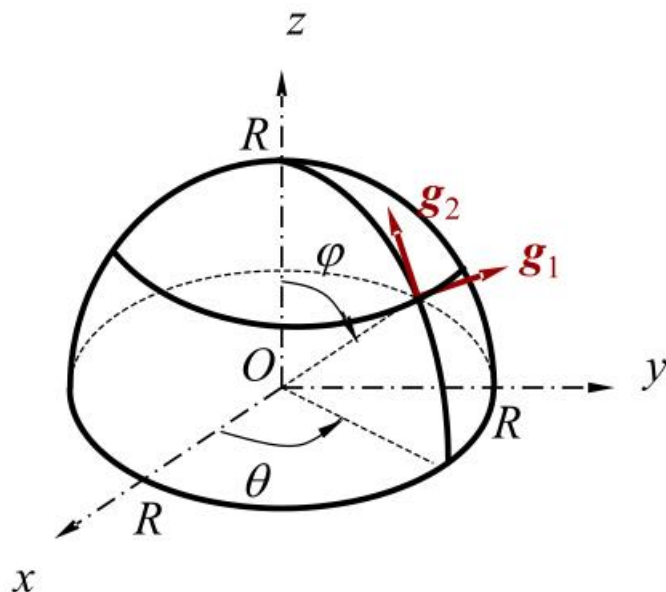
$$= \frac{H^2}{R^2 [H^2 + (R + C \cos \theta)^2] z^2} \begin{bmatrix} H^2 + R^2 + C^2 + 2RC \cos \theta & RCz \sin \theta \\ RCz \sin \theta & R^2 z^2 \end{bmatrix}$$

$$\bullet \frac{1}{H} \begin{bmatrix} -Rz \sin \theta & Rz \cos \theta & 0 \\ C + R \cos \theta & R \sin \theta & H \end{bmatrix} \begin{Bmatrix} \mathbf{i} \\ \mathbf{j} \\ \mathbf{k} \end{Bmatrix}$$

$$= \frac{H}{R^2 [H^2 + (R + C \cos \theta)^2] z^2} \begin{bmatrix} -R(H^2 + R^2 + RC \cos \theta)z \sin \theta \\ R^2(R + C \cos \theta)z^2 \cos \theta \end{bmatrix}$$

$$\begin{bmatrix} R(H^2 + R^2 + C^2)z \cos \theta + R^2 Cz(1 + \cos^2 \theta) & HRCz \sin \theta \\ R^2(R + C \cos \theta)z^2 \sin \theta & HR^2 z^2 \end{bmatrix} \begin{Bmatrix} \mathbf{i} \\ \mathbf{j} \\ \mathbf{k} \end{Bmatrix}$$

1.15 二维空间为半径为 R 的半球面，如图， $x^1=\theta$ ， $x^2=\varphi$ 。
用两种方法求 $\mathbf{g}_\alpha$ ， $\mathbf{g}^\beta$ ， $g_{\alpha\beta}$ ， $g^{\alpha\beta}$ ($\alpha, \beta=1, 2$)。



解：

$$\begin{aligned}\mathbf{r} &= x^3 \sin x^1 \cos x^2 \mathbf{i} + x^3 \sin x^1 \sin x^2 \mathbf{j} + x^3 \cos x^1 \mathbf{k} \\ &= R \sin \theta \cos \varphi \mathbf{i} + R \sin \theta \sin \varphi \mathbf{j} + R \cos \theta \mathbf{k}\end{aligned}$$

$$\begin{Bmatrix} \mathbf{g}_1 \\ \mathbf{g}_2 \end{Bmatrix} = \begin{bmatrix} \frac{\partial x^{1'}}{\partial x^1} & \frac{\partial x^{2'}}{\partial x^1} & \frac{\partial x^{3'}}{\partial x^1} \\ \frac{\partial x^{1'}}{\partial x^2} & \frac{\partial x^{2'}}{\partial x^2} & \frac{\partial x^{3'}}{\partial x^2} \end{bmatrix} \begin{Bmatrix} \mathbf{i} \\ \mathbf{j} \\ \mathbf{k} \end{Bmatrix}$$

$$\begin{Bmatrix} \mathbf{g}_1 \\ \mathbf{g}_2 \end{Bmatrix} = R \begin{bmatrix} \cos \theta \cos \varphi & \cos \theta \sin \varphi & -\sin \theta \\ -\sin \theta \sin \varphi & \sin \theta \cos \varphi & 0 \end{bmatrix} \begin{Bmatrix} \mathbf{i} \\ \mathbf{j} \\ \mathbf{k} \end{Bmatrix}$$

$$[g_{\alpha\beta}] = [\mathbf{g}_\alpha \bullet \mathbf{g}_\beta]$$

$$= R^2 \begin{bmatrix} \cos \theta \cos \varphi & \cos \theta \sin \varphi & -\sin \theta \\ -\sin \theta \sin \varphi & \sin \theta \cos \varphi & 0 \end{bmatrix} \begin{bmatrix} \cos \theta \cos \varphi & -\sin \theta \sin \varphi \\ \cos \theta \sin \varphi & \sin \theta \cos \varphi \\ -\sin \theta & 0 \end{bmatrix}$$

$$= R^2 \begin{bmatrix} 1 & 0 \\ 0 & \sin^2 \theta \end{bmatrix}$$

$$[g^{\alpha\beta}] = [g_{\alpha\beta}]^{-1} = \frac{1}{R^2} \begin{bmatrix} 1 & 0 \\ 0 & 1/\sin^2 \theta \end{bmatrix}$$

$$\{\mathbf{g}^\beta\} = [g^{\alpha\beta}]\{\mathbf{g}_\alpha\}$$

$$= \frac{1}{R^2} \begin{bmatrix} 1 & 0 \\ 0 & 1/\sin^2 \theta \end{bmatrix} R \begin{bmatrix} \cos \theta \cos \varphi & \cos \theta \sin \varphi & -\sin \theta \\ -\sin \theta \sin \varphi & \sin \theta \cos \varphi & 0 \end{bmatrix} \begin{Bmatrix} \mathbf{i} \\ \mathbf{j} \\ \mathbf{k} \end{Bmatrix}$$

$$= \frac{1}{R} \begin{bmatrix} \cos \theta \cos \varphi & \cos \theta \sin \varphi & -\sin \theta \\ -\sin \varphi / \sin \theta & \cos \varphi / \sin \theta & 0 \end{bmatrix} \begin{Bmatrix} \mathbf{i} \\ \mathbf{j} \\ \mathbf{k} \end{Bmatrix}$$

1.16 已知：圆柱坐标系中、球坐标系中矢量的逆变分量 v^i 。
利用题 1.13 结果分别求两个坐标系中的协变分量 v_i 。

解： $\{v_i\} = [g_{ij}]\{v^j\}$

(1) 圆柱坐标系
$$\begin{Bmatrix} v_1 \\ v_2 \\ v_3 \end{Bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} v^1 \\ v^2 \\ v^3 \end{Bmatrix} = \begin{Bmatrix} v^1 \\ r^2 v^2 \\ v^3 \end{Bmatrix}$$

(2) 球坐标系
$$\begin{Bmatrix} v_1 \\ v_2 \\ v_3 \end{Bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & (r \sin \theta)^2 \end{bmatrix} \begin{Bmatrix} v^1 \\ v^2 \\ v^3 \end{Bmatrix} = \begin{Bmatrix} v^1 \\ r^2 v^2 \\ (r \sin \theta)^2 v^3 \end{Bmatrix}$$

1.17 求：题 1.13 所示圆柱坐标和球坐标 x^i ，与笛卡儿坐标 $x^{j'}$ 的转换系数 $\beta_{j'}^i$ 与 $\beta_i^{j'}$ 。

解：
$$[\beta_{i}^{j'}] = \left[\frac{\partial x^{j'}}{\partial x^i} \right]_{ij}, \quad [\beta_{j'}^i] = [\beta_i^{j'}]^{-1}$$

圆柱坐标系：
$$\begin{aligned} x^{1'} &= x^1 \cos x^2 \\ x^{2'} &= x^1 \sin x^2 \\ x^{3'} &= x^3 \end{aligned}$$

$$[\beta_{i}^{j'}] = \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -r \sin \theta & r \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad [\beta_{j'}^i] = \begin{bmatrix} \cos \theta & -\sin \theta / r & 0 \\ \sin \theta & \cos \theta / r & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

球坐标系:

$$x^{1'} = x^1 \sin x^2 \cos x^3$$

$$x^{2'} = x^1 \sin x^2 \sin x^3$$

$$x^{3'} = x^1 \cos x^2$$

$$[\beta_i^{j'}] = \begin{bmatrix} \sin \theta \cos \varphi & \sin \theta \sin \varphi & \cos \theta \\ r \cos \theta \cos \varphi & r \cos \theta \sin \varphi & -r \sin \theta \\ -r \sin \theta \sin \varphi & r \sin \theta \cos \varphi & 0 \end{bmatrix}$$

$$[\beta_{j'}^i] = \begin{bmatrix} \sin \theta \cos \varphi & \cos \theta \cos \varphi / r & -\sin \varphi / (r \sin \theta) \\ \sin \theta \sin \varphi & \cos \theta \sin \varphi / r & \cos \varphi / (r \sin \theta) \\ \cos \theta & -\sin \theta / r & 0 \end{bmatrix}$$

1.18 (1) 已知:笛卡儿坐标系中 $\boldsymbol{v}$ 的分量为 $v^{1'}$, $v^{2'}$, $v^{3'}$;
求:圆柱坐标中 $\boldsymbol{v}$ 的分量 v^1 , v^2 , v^3 。

(2) 已知:笛卡儿坐标系中 $\boldsymbol{v}$ 的分量为 $v_{1'}$, $v_{2'}$, $v_{3'}$;
求:球坐标中 $\boldsymbol{v}$ 的分量 v_1 , v_2 , v_3 。

解: $\{\boldsymbol{v}^i\} = [\beta_{j'}^i] \{\boldsymbol{v}^{j'}\}, \quad \{\boldsymbol{v}_i\} = [\beta_i^{j'}] \{\boldsymbol{v}_{j'}\}$

$$(1) \quad \begin{Bmatrix} v^1 \\ v^2 \\ v^3 \end{Bmatrix} = \begin{bmatrix} \beta_{1'}^1 & \beta_{2'}^1 & \beta_{3'}^1 \\ \beta_{1'}^2 & \beta_{2'}^2 & \beta_{3'}^2 \\ \beta_{1'}^3 & \beta_{2'}^3 & \beta_{3'}^3 \end{bmatrix} \begin{Bmatrix} v^{1'} \\ v^{2'} \\ v^{3'} \end{Bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta / r & \cos \theta / r & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} v^{1'} \\ v^{2'} \\ v^{3'} \end{Bmatrix}$$

$$(2) \quad \begin{Bmatrix} v_1 \\ v_2 \\ v_3 \end{Bmatrix} = \begin{bmatrix} \beta_1^{1'} & \beta_1^{2'} & \beta_1^{3'} \\ \beta_2^{1'} & \beta_2^{2'} & \beta_2^{3'} \\ \beta_3^{1'} & \beta_3^{2'} & \beta_3^{3'} \end{bmatrix} \begin{Bmatrix} v^{1'} \\ v^{2'} \\ v^{3'} \end{Bmatrix} = \begin{bmatrix} \sin \theta \cos \varphi & \sin \theta \sin \varphi & \cos \theta \\ r \cos \theta \cos \varphi & r \cos \theta \sin \varphi & -r \sin \theta \\ -r \sin \theta \sin \varphi & r \sin \theta \cos \varphi & 0 \end{bmatrix} \begin{Bmatrix} v^{1'} \\ v^{2'} \\ v^{3'} \end{Bmatrix}$$

1.27 设一动点轨迹为 $x^i(t)$ ($t \geq 0$, 标量), 定义

$$v^i = \lim_{\Delta t \rightarrow 0} \frac{x^i(t + \Delta t) - x^i(t)}{\Delta t} = \frac{dx^i}{dt}。$$

求证: v^i 为矢量分量。

1.28 由应变 ε_{ij} 的定义 $d\hat{\boldsymbol{r}} \bullet d\hat{\boldsymbol{r}} - d\boldsymbol{r} \bullet d\boldsymbol{r} = 2\varepsilon_{ij}dx^i dx^j$ 出发, 求证: ε_{ij} 是对称二阶张量的分量。式中 dx^i 是介质的拉格朗日坐标的微分。

1.38 在笛卡儿坐标系中，各向同性材料的弹性关系为

$$\begin{aligned}\varepsilon_{11} &= \frac{1}{E} [\sigma^{11} - \nu(\sigma^{22} + \sigma^{33})], & \varepsilon_{12} &= \frac{1+\nu}{E} \sigma^{12} \\ \varepsilon_{22} &= \frac{1}{E} [\sigma^{22} - \nu(\sigma^{33} + \sigma^{11})], & \varepsilon_{23} &= \frac{1+\nu}{E} \sigma^{23} \\ \varepsilon_{33} &= \frac{1}{E} [\sigma^{33} - \nu(\sigma^{11} + \sigma^{22})], & \varepsilon_{31} &= \frac{1+\nu}{E} \sigma^{31}\end{aligned}$$

(1) 利用商法则证明此式必定可以表示为一个张量的代数运算等式，写出其实体形式，说明等式中各阶张量的阶数。

(2) 将上式表示为可运用于任意坐标系的张量分量形式。

(3) 写出任意坐标系中的协变分量 D_{ijkl} 用 E , ν 及度量张量分量表达的形式，以及 D 的并矢表达式。

解: (1)

$$\varepsilon_{ij} = -\frac{\nu}{E} \sigma^k_{\bullet k} g_{ij} + \frac{1+\nu}{E} \sigma_{ij}$$

$$\boldsymbol{\varepsilon} = -\frac{\nu}{E} \sigma^k_{\bullet k} \mathbf{G} + \frac{1+\nu}{E} \boldsymbol{\sigma} = \mathbf{D} : \boldsymbol{\sigma}$$

(2)

$$\varepsilon_{ij} = -\frac{\nu}{E} \sigma^k_{\bullet k} g_{ij} + \frac{1+\nu}{E} \sigma_{ij}$$

$$= -\frac{\nu}{E} \sigma^k_{\bullet k} g_{ij} + \frac{1+\nu}{2E} (\sigma_{ij} + \sigma_{ji})$$

$$= -\frac{\nu}{E} \sigma^{kl} g_{ij} g_{kl} + \frac{1+\nu}{2E} (g_{ik} g_{jl} + g_{jk} g_{il}) \sigma^{kl}$$

$$= D_{ijkl} \sigma^{kl}$$

$$(3) \quad D_{ijkl} = -\frac{\nu}{E} g_{ij} g_{kl} + \frac{1+\nu}{2E} (g_{ik} g_{jl} + g_{il} g_{jk})$$

$$\boldsymbol{D} = -\frac{\nu}{E} \boldsymbol{G} \boldsymbol{G} + \frac{1+\nu}{E} \boldsymbol{I}$$

$$\boldsymbol{I} = \frac{1}{2} (g_{ik} g_{jl} + g_{il} g_{jk}) \boldsymbol{g}^i \boldsymbol{g}^j \boldsymbol{g}^k \boldsymbol{g}^l$$

1.39 已知：矩阵 $[A]$, $[B]$, $[C]=[A][B]$, $a=\det[A]$, $b=\det[B]$, $c=\det[C]$ 。求：利用置换符号证明： $c=ab$ 。

证明： $c_{\bullet j}^i = a_{\bullet k}^i b_{\bullet j}^k$

$$\begin{aligned} c &= c_{\bullet r}^1 c_{\bullet s}^2 c_{\bullet t}^3 e^{rst} = a_{\bullet l}^1 b_{\bullet r}^l a_{\bullet m}^2 b_{\bullet s}^m a_{\bullet n}^3 b_{\bullet t}^n e^{rst} \\ &= a_{\bullet l}^1 a_{\bullet m}^2 a_{\bullet n}^3 b e^{lmn} = ab \end{aligned}$$

或

$$\begin{aligned} c &= c_{\bullet 1}^r c_{\bullet 2}^s c_{\bullet 3}^t e_{rst} = a_{\bullet l}^r b_{\bullet 1}^l a_{\bullet m}^s b_{\bullet 2}^m a_{\bullet n}^t b_{\bullet 3}^n e_{rst} \\ &= b_{\bullet 1}^l b_{\bullet 2}^m b_{\bullet 3}^n a e_{lmn} = ab \end{aligned}$$

1.40 已知：矩阵 $[a_{\bullet j}^i]$ 中某两列的元素成比例，例如：

$a_{\bullet 1}^i = k a_{\bullet 2}^i$ k 为一个实数。求：利用置换符号证明：

$$\det(a_{\bullet j}^i) = 0。$$

证明：

$$\begin{aligned}\det(a_{\bullet j}^i) &= a_{\bullet 1}^l a_{\bullet 2}^m a_{\bullet 3}^n e_{lmn} = k a_{\bullet 2}^l a_{\bullet 2}^m a_{\bullet 3}^n e_{lmn} \\ &= -k a_{\bullet 2}^l a_{\bullet 2}^m a_{\bullet 3}^n e_{mln} = -\det(a_{\bullet j}^i)\end{aligned}$$

$$\therefore \det(a_{\bullet j}^i) = 0$$

1.41 质量为 m ，绕定点 O 以角速度 ω 转动的质点（如图），其动量矩矢量的定义为 $L = m\mathbf{r} \times \mathbf{v}$ ，其中， $\mathbf{r}$ 为定点 O 至质点的矢径， $\mathbf{v}$ 为质点的线速度。

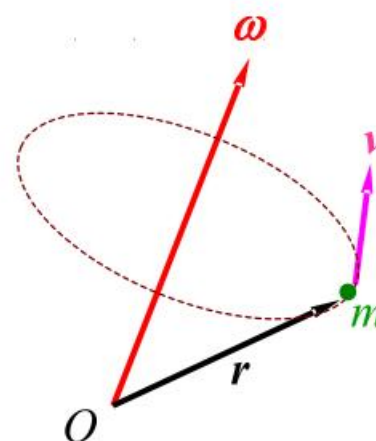
求证： $L = I \cdot \omega$ ，式中 I 为惯性矩张量， $I = m[(\mathbf{r} \cdot \mathbf{r})\mathbf{G} - \mathbf{r}\mathbf{r}]$

证明：

$$\begin{aligned} \mathbf{L} &= m\mathbf{r} \times \mathbf{v} = m\mathbf{r} \times (\boldsymbol{\omega} \times \mathbf{r}) \\ &= m[\boldsymbol{\omega}(\mathbf{r} \cdot \mathbf{r}) - \mathbf{r}(\mathbf{r} \cdot \boldsymbol{\omega})] \end{aligned}$$

$$\begin{aligned} L^i &= m[\omega^i r^m r_m - r^i r_k \omega^k] \\ &= m[\delta_k^i r^m r_m - r^i r_k] \omega^k \\ &= I_{\bullet k}^i \omega^k \end{aligned}$$

$$\mathbf{L} = m[(\mathbf{r} \cdot \mathbf{r})\mathbf{G} - \mathbf{r}\mathbf{r}] \bullet \boldsymbol{\omega} = \mathbf{I} \bullet \boldsymbol{\omega}$$



1.42 求图1.11所示球坐标系中的面元矢量 da^1 , da^2 , da^3 。

解: $da^i = ds^j \times ds^k$

$$\begin{aligned} \begin{Bmatrix} ds^1 \\ ds^2 \\ ds^3 \end{Bmatrix} &= \begin{bmatrix} g^{11} & g^{12} & g^{13} \\ g^{21} & g^{22} & g^{23} \\ g^{31} & g^{32} & g^{33} \end{bmatrix} \begin{Bmatrix} ds_1 \\ ds_2 \\ ds_3 \end{Bmatrix} \\ &= \begin{bmatrix} g^{11} & g^{12} & g^{13} \\ g^{21} & g^{22} & g^{23} \\ g^{31} & g^{32} & g^{33} \end{bmatrix} \begin{bmatrix} dx^1 & 0 & 0 \\ 0 & dx^2 & 0 \\ 0 & 0 & dx^3 \end{bmatrix} \begin{Bmatrix} \mathbf{g}_1 \\ \mathbf{g}_2 \\ \mathbf{g}_3 \end{Bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/r^2 & 0 \\ 0 & 0 & 1/(r \sin \theta)^2 \end{bmatrix} \begin{bmatrix} dx^1 & 0 & 0 \\ 0 & dx^2 & 0 \\ 0 & 0 & dx^3 \end{bmatrix} \begin{Bmatrix} \mathbf{g}_1 \\ \mathbf{g}_2 \\ \mathbf{g}_3 \end{Bmatrix} = \begin{Bmatrix} \mathbf{g}_1 dx^1 \\ \mathbf{g}_2 dx^2 / r^2 \\ \mathbf{g}_3 dx^3 / (r \sin \theta)^2 \end{Bmatrix} \end{aligned}$$

$$d\boldsymbol{a}^1 = d\boldsymbol{s}^2 \times d\boldsymbol{s}^3 = \boldsymbol{g}_2 \frac{dx^2}{r^2} \times \boldsymbol{g}_3 \frac{dx^3}{(r \sin \theta)^2} = \boldsymbol{g}_1 \frac{dx^2 dx^3}{(r^2 \sin \theta)^2}$$

$$d\boldsymbol{a}^2 = d\boldsymbol{s}^3 \times d\boldsymbol{s}^1 = \boldsymbol{g}_3 \frac{dx^3}{(r \sin \theta)^2} \times \boldsymbol{g}_1 dx^1 = \boldsymbol{g}_2 \frac{dx^3 dx^1}{(r \sin \theta)^2}$$

$$d\boldsymbol{a}^3 = d\boldsymbol{s}^1 \times d\boldsymbol{s}^2 = \boldsymbol{g}_1 dx^1 \times \boldsymbol{g}_2 \frac{dx^2}{r^2} = \boldsymbol{g}_3 \frac{dx^1 dx^2}{r^2}$$